

Problem Set - 8

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1 Mechanics

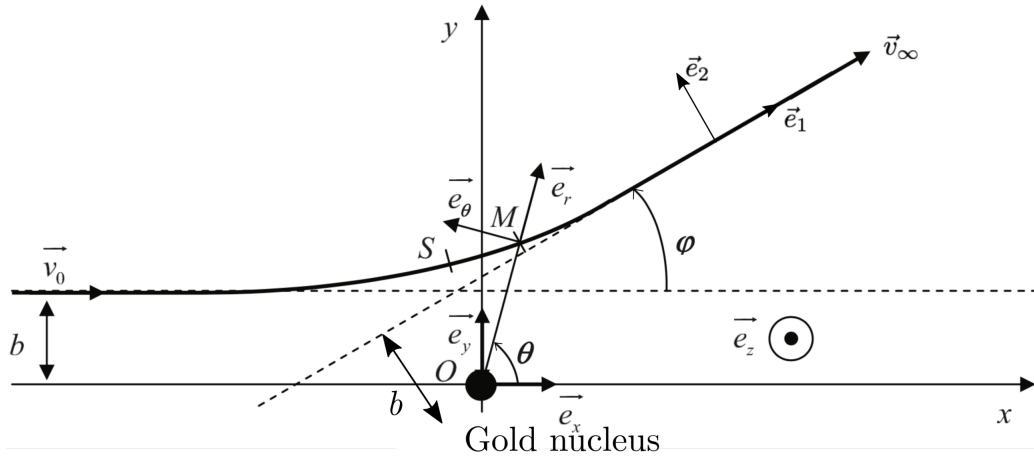
1.1 Beer Can

A space station is in a circular orbit about the Earth at a radius R . An astronaut on a space walk happens to be a distance R on the far side of the station on the line joining the station to the center of the Earth. With practice, the astronaut can throw a beer can so that it appears to orbit the space station in the plane of the motion of the space station about the Earth according to an observer on the station. You may disregard the gravitational attraction between the beer can and the space station. (A Guide to Physics Problems, 1.44)

- In what direction and with what velocity relative to the station should the beer can be thrown?
- What is the period, size and shape of the beer can's orbit, relative to the space station?

1.2 The Rutherford Experiment

In 1909 and the following years, Ernest Rutherford, conducted a series of experiments that contributed to the understanding of the nature of atoms [2, 3]. In these experiments, a beam of alpha particles (helium atoms stripped of two electrons ${}^4\text{He}^{2+}$ - positively charged - obtained from the radioactive decomposition of radium) is sent on a thin sheet of gold. He observed that some alpha particles were going through the gold sheet with various deflection angles and others "bounced" on it. This observation observations could only be explained by assuming that the gold atoms are constituted of a positively charged nucleus of small dimensions (compared with the atom size) surrounded by a cloud of negatively charged electrons, therefore supporting this conception of the structure of matter. The objective of this problem is to determine the deflection angle of an alpha particle as it interacts with a gold nucleus (both being positively charged), a problem addressed by Rutherford in 1911 [5].



A gold nucleus represented by a point particle is located at the origin O of a orthonormal Cartesian referential $(O, \vec{e}_x, \vec{e}_y, \vec{e}_z)$.

- At time $t = 0$, the alpha particle comes from "infinity" with a rectilinear uniform motion characterized by velocity vector $\vec{v}_0 = v_0 \vec{e}_x$. The distance from point O to the trajectory of the particle as it comes from infinity is denoted by b .
- During deflection, the particle is located at the point $M(t)$ with polar coordinates $(r(t), \theta(t))$ and travels with velocity and acceleration $\vec{v}(t)$ and $\vec{a}(t)$. We denote by B the local cylindrical basis $(\vec{e}_r, \vec{e}_\theta, \vec{e}_z)$ attached to point M .
- After deflection, the particle leaves at $t \rightarrow +\infty$ with a constant velocity vector $\vec{v}_\infty$ that makes an angle φ with the (Ox) direction. We define the orthonormal Cartesian basis $(\vec{e}_1, \vec{e}_2, \vec{e}_z)$ with $\vec{e}_1$ aligned with $\vec{v}_\infty$.

The force exerted by the nucleus on the particle is a repulsive electrostatic force of the type $\vec{F} = K\vec{r}/r^3$ with $\vec{r} = \overrightarrow{OM}$ and $r = |\vec{r}|$ and $K > 0$. Newton's second law of motion relates the acceleration $\vec{a}$ of the particle to the force $\vec{F}$ it is subjected to through $\vec{F} = m\vec{a}$. We define the angular momentum vector $\vec{L} = \vec{r} \times \vec{v}$ and the Laplace-Runge-Lenz vector $\vec{A} = \vec{v} \times \vec{L} + K\vec{r}/mr$. (2024 Calculus Final Exam, École Polytechnique)

- a) Show that $\vec{L}$ and $\vec{A}$ are constant vectors.
- b) Calculate $\vec{A}(t = 0)$ and $\vec{A}(t \rightarrow +\infty)$.
- c) Calculate the deflection angle φ .
- d) What are the values of the deflection angle for $b \rightarrow 0$ and $b \rightarrow +\infty$. For what values of b do the alpha particle "bounces" on the gold sheet ($\varphi \geq \pi/2$)?

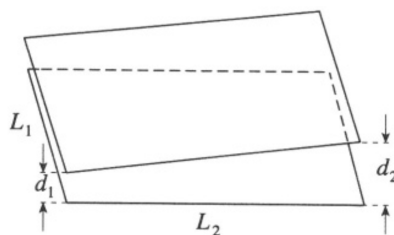
2 Electromagnetism

2.1 Not-so-parallel Plate Capacitor

A capacitor is formed by two rectangular conducting plates having edges L_1 and L_2 . The plates are not parallel. One pair of edges of length L_1 is separated by a distance d_1 everywhere, and the other pair of edges of length L_2 is separated by d_2 everywhere; $d_2 > d_1$. (A Guide to Physics Problems, 3.16)

a) Neglecting edge effects, when a voltage difference V is placed across the two conductors, find the potential everywhere between the plates.

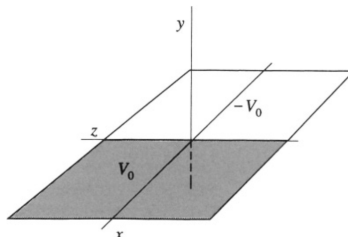
b) Determine the capacitance.



2.2 Applications of the Method of Conformal Mapping

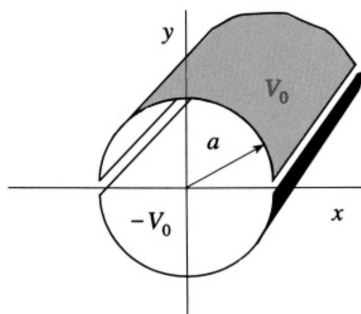
2.2.1 Potential above Half Planes

An infinite conducting plane is divided by the line $z = 0$. For $x > 0$ the potential in the plane is $+V_0$ while for $x < 0$ the potential is $-V_0$. Evaluate the potential everywhere. (A Guide to Physics Problems, 3.25)



2.2.2 Potential of Halved Cylinder

Consider an infinitely long conducting cylinder of radius a with its axis coinciding with the z -axis. One half of the cylinder (cut the long way) ($y > 0$) is kept at a constant potential $+V_0$, while the other half ($y < 0$) is kept at a constant potential $-V_0$. Find the potential for all points inside the cylinder and the field $\vec{E}$ along the z -axis. (A Guide to Physics Problems, 3.26)

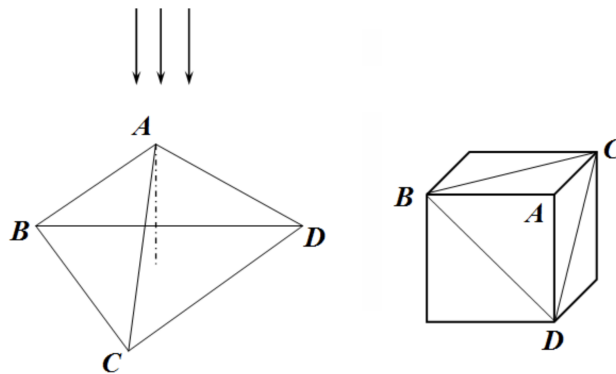


3 Optics

3.1 Refractive Pyramid

A regular triangular pyramid $ABCD$ is made of a transparent material with a refractive index of $n = 1.6$. All angles at the top of the pyramid A are right angles. At the base of the pyramid lies a regular triangle BCD . This pyramid can also be represented as a “corner cut off from a cube”, whose side lengths are equal to $a = 2.0$ mm. The pyramid is uniformly illuminated so that the light falls along its height, perpendicular to the plane of the base. (Zhautykov 2013, P1.C)

- 1) Draw the area on the base of the pyramid illuminated by the light refracted by face ABC .
- 2) A screen is located parallel to the base of the pyramid at a distance of $L = 10$ cm from it. Draw the areas illuminated by the light refracted by the pyramid on this screen. Indicate the position and dimensions of these areas.



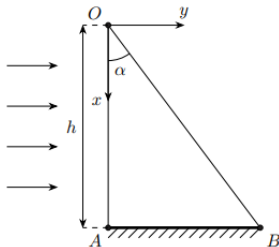
3.2 Cassini Division

In an opaque flat layer of thickness h a hole is drilled, the shape of the side surface of which is a truncated cone with base diameters d and D , the axis of which is perpendicular to the flat surfaces of the layer. The side surface of the cone is silvered and perfectly reflects the light falling on it. Behind the layer at a distance L from the nearest flat surface of the layer there is a flat screen. (RuPhO 2023)

- 1) Draw a picture of what you see on the screen. Indicate all the characteristic geometric dimensions in the drawing.
- 2) How will the picture from point 1 change if a thin collecting lens is placed between the layer and the screen? The focal length of the lens $F = L/2$, the main optical axis of the lens coincides with the axis of the cone, the screen is located in the focal plane of the lens.

3.3 Inhomogeneity in a Prism

Experimenter Gluck conducts optical studies with a solid prism, cross section OAB which is a right triangle with legs AB and $OA = h$. All faces of the prism are either parallel or perpendicular to the plane of the drawing.



If we introduce a rectangular coordinate system (x, y) with origin at point O as shown in the figure, the refractive index of the prism material depends only on the coordinate x as:

$$n(x) = \frac{3}{2 - x/h}$$

Gluck decided to completely illuminate the face of the prism containing OA with a light beam perpendicular to it. The bottom of the prism, containing AB is covered with a substance that completely absorbs the light that hits it. The prism is located in air, the refractive index of which can be considered equal to one. (RuPhO 2024)

- 1) Let us consider a ray entering a prism at a point with coordinate x_0 . Find the equation of the trajectory of this ray inside the prism before it hits the sides AB and OB .
- 2) At what angle values $\angle AOB = \alpha$ do the rays of light hit all points on the side OB prism sections?
- 3) At what angle values α all rays that reached the side OB sections of a prism are refracted and pass into the air?

4 Miscellaneous

4.1 Gyromagnetic Constant

a) A solid sphere of radius r , mass m and homogeneously distributed charge q is rotating around itself with angular velocity ω . Calculate the ratio γ of its dipole moment to its angular momentum.

b) For an object with mass m , homogeneously distributed charge q and unknown shape, calculate γ .

(Eylül Kamp Sonu Sınavı, ???)

4.2 Metamaterials

Metamaterials are composite materials whose properties are due not so much to the properties of its constituent elements but due to artificially tailored periodic structures. Metamaterials are synthesized in modern nanolaboratories by implantation different periodic structures with a variety of geometric shapes into the original natural material, which substantially modifies its physical properties. In a very rough approximation, those implants can be treated as artificially made atoms of extremely large size immersed into the original material. While synthesizing the metamaterial developer has the opportunity of varying various free parameters (structure sizes and constant or varying period between them, etc.). In one nanolaboratory the metamaterial has been manufactured in the form of a wire of the length $L = 5.00$ cm and radius $R = 1.00$ mm, whose conductivity depends on the distance from its axis according to the law $\sigma_0 = \beta r$.

(Zhautykov 2014, P3)

Physical properties of the wire have been experimentally measured and are presented in the following table:

PHYSICAL PROPERTY	NUMERICAL VALUE
Conductivity $\sigma_0 = \beta r$	$\beta = 1.00 \times 10^9 S / m^2$
Heat transfer coefficient	$\alpha = 20 W / (m^2 \cdot K)$
Thermal conductivity coefficient	$\kappa = 0,01 W / (m \cdot K)$
Young's modulus	$E = 1.00 \times 10^7 Pa$
Linear expansion coefficient	$\gamma = 1.00 \times 10^{-6} K^{-1}$

1) Find an analytic formula for the total resistance R_0 of the wire, and calculate its numerical value.

An electric current $I = 1A$ is made to pass through the wire. It is known that the heat exchange with the environment obeys the Newton-Richman law,

$$P_{ext} = \alpha(T_s - T_0)$$

where P_{ext} stands for the power loss per unit surface of the wire with the surface temperature T_s , $T_0 = 293K$ denotes the ambient temperature and α is a constant, called the heat transfer coefficient.

2) Find an analytic formula for the surface temperature T_s of the wire and calculate its numerical value.

The wire temperature varies with the depth due to the phenomenon known as thermal conductivity, which is described by the Fourier law

$$P = -\kappa S \frac{\Delta T}{\Delta x}$$

where P designates the power of the heat flow between the opposite faces of the parallelepiped with the square S , ΔT is the temperature difference between the faces of the parallelepiped situated at a distance Δx from each other, and κ is called the heat transfer coefficient.

3) Find an analytic formula for the temperature T_{max} in the center of the wire, and calculate its numerical value.

4) Find an analytic formula for the change δR_T of the wire radius due to its thermal expansion and calculate its numerical value.

Attention! In all further calculations assume that the wire is infinitely long. **5)** Find an analytic formula for the magnetic induction inside the wire as a function of the distance r from its axis.

6) Find an analytic formula for the energy of the magnetic field inside the wire, and calculate its numerical value.

7) The electric current causes an appearance of mechanical stress in the wire. Find an analytic formula for the pressure $p(r)$ inside the wire as a function of the distance r from its axis.

8) Find an analytic formula for the mechanical stress energy W_σ of the wire, and calculate its numerical value.

9) Find an analytic formula for the change δR_σ of the wire radius due to its mechanical stress, and calculate its numerical value.

10) Find the value of the thermal expansion coefficient γ such that the total change of the wire radius would be zero when an electric current was passing through it.